Keng-Wit Lim

Education

California Institute of Technology, Applied Mechanics — Pasadena, CA, 2010 – 2014

North Carolina State University, Structural Engineering and Mechanics — Raleigh, NC, 2001 – 2003

University of Wales, Civil Engineering — Swansea, UK, 1996 – 1999

Experience

Simpson Gumpertz & Heger, Senior Technical Manager, Structural Mechanics and Materials — Los Angeles, CA, 2023 – present, 3 years

Assessment of light and heavy rail vehicle crashworthiness, crash energy management system conceptual design.

Simpson Gumpertz & Heger, Senior Consulting Engineer — Los Angeles, CA, 2015 – 2023, 8 years

Structural optimization, design and failure analysis of engineering systems - precision structures, geomechanical, energy, transportation, defense.

California Institute of Technology, Postdoctoral Fellow — Pasadena, CA, 2014 – 2015, 1 year

- Computational methods for fragmentation and comminution of particulate media.
- Abaqus UEL/UMAT in C++ and Fortran, implementation of object-oriented nonlinear finite elements for hyperelasticity and elasto-plasticity.

Simpson Gumpertz & Heger, Staff I and II — Waltham, MA, 2004 – 2010, 6 years

- Soil-structure interaction and finite element analyses of energy-related facilities
- weapons-grade plutonium conversion and vitrification (for peaceful nuclear fuel), radioactive waste treatment units, power generation units.
- Design, field investigation, risk analysis, and determination of repair priorities of pipelines.
- Modeling components for rail vehicle crash-worthiness protection. Structural field support and testing, failure investigations.

Interests

Computational Solid Mechanics

Languages

English
Native or bilingual proficiency

Bahasa Malaysia
Native or bilingual proficiency

Mandarin
Elementary proficiency